

Mrs Cooper
Mrs Greenberg
Ms Lau

Mr Feng
Mrs Kerr
Mrs Squires

Name:

Teacher:.....



Pymble Ladies' College

MATHEMATICS EXTENSION 1

Year 11 ASSESSMENT TASK 2

Term 2 2023

General**Instructions**

- Working time – 55 minutes
- Write using black or blue pen
- Erasable pens are not to be used
- Calculators approved by NESA may be used
- Reference sheets are provided

Total marks

33

SECTION 1 – 5 marks

- Attempt Questions 1-5

SECTION II – 28 marks

- Attempt all Questions
- Answer each question in the appropriate space in the Answer Booklet.

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SECTION I

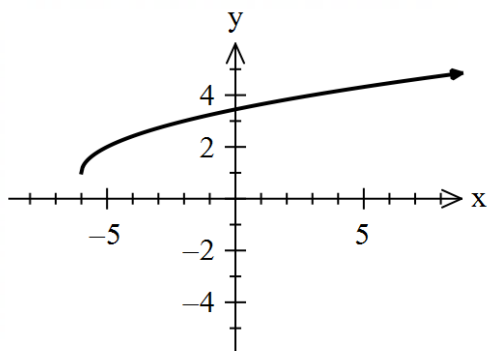
Multiple Choice (5 marks)

Answer using the Multiple-Choice Answer Sheet.

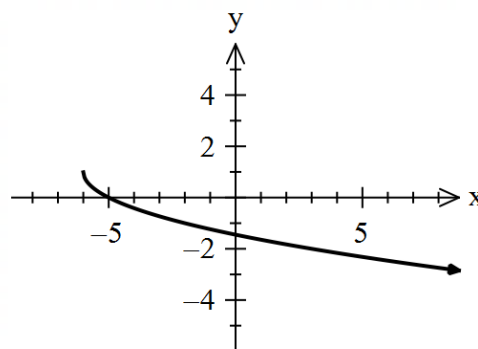
- 1 A function is defined as $f(x) = x^2 - 2x - 5$ over the domain $x \leq 1$.

Which of the following is the graph of $f^{-1}(x)$?

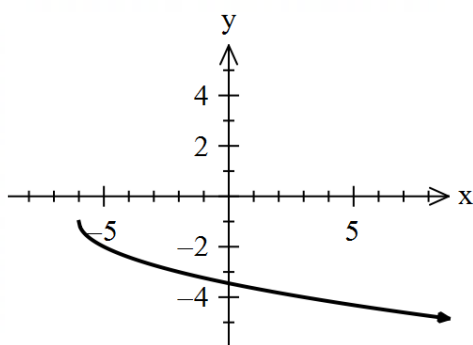
(A)



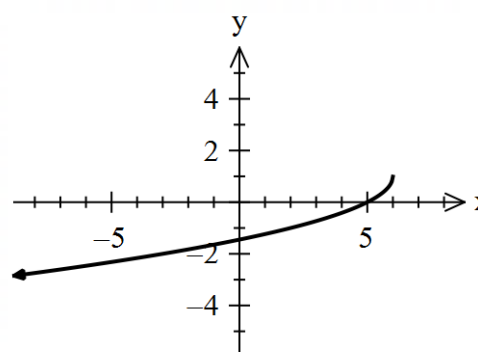
(B)



(C)



(D)



- 2 The function $f(x) = 96 \cos(3x) \sin(3x)$ can be written as $f(x) = a \sin(bx)$.

What are the values of a and b ?

(A) $a = 48, b = 3$

(B) $a = 48, b = 6$

(C) $a = 192, b = 3$

(D) $a = 192, b = 6$

3 What is the exact value of $\sin\left(2\cos^{-1}\left(\frac{-\sqrt{15}}{8}\right)\right)$?

(A) $\frac{7}{8}$

(B) $\frac{-7}{8}$

(C) $\frac{7\sqrt{15}}{32}$

(D) $\frac{-7\sqrt{15}}{32}$

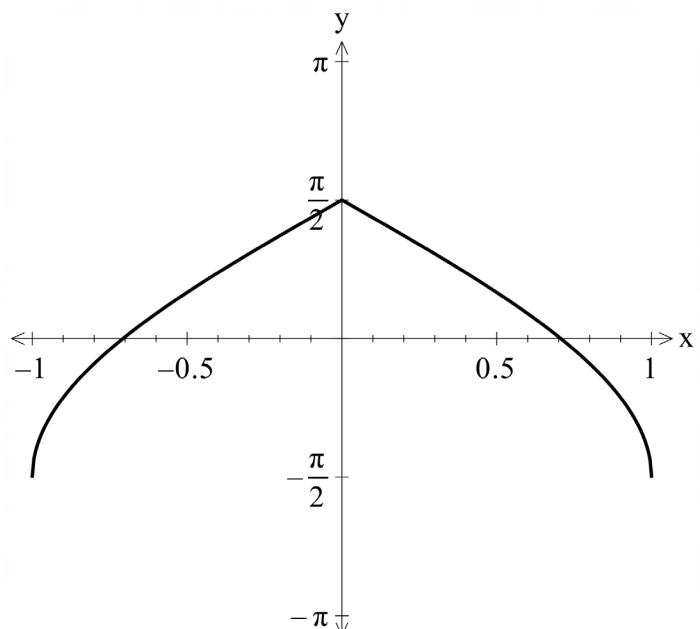
4 What is the equation of the function shown in the graph below?

(A) $y = |2\sin^{-1}(x)|$

(B) $y = \left|2\cos^{-1}(x) - \frac{\pi}{2}\right|$

(C) $y = 2\cos^{-1}(|x|) - \frac{\pi}{2}$

(D) $y = 2\sin^{-1}(|x|)$



5 A curve is defined by the parametric equations $x = \cos 2\theta$ and $y = \sin \theta \cos \theta$. Which of the following represents the Cartesian equation?

(A) $x^2 + y^2 = 1$

(B) $x^2 + 4y^2 = 1$

(C) $x^2 + 2y^2 = 1$

(D) $x^2 + y^2 = 4$

SECTION II**25 Marks**

Answer these questions in the Answer Book provided

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (10 marks)	Marks
(a) Find the exact value of the following, showing all necessary working.	
(i) $\arccos\left(-\frac{\sqrt{3}}{2}\right)$	1
(ii) $\cos\left(\tan^{-1}(-1)\right)$	2
(iii) $\sin^{-1}\left(\sin\frac{9\pi}{4}\right)$	2
(b) Sketch the graph expressed parametrically by $(p^2, 3)$.	1
(c) If $t = \tan \theta$, express $\frac{\cot 2\theta}{1 - \operatorname{cosec} 2\theta}$ in terms of t in simplest form.	2
(d) Find the exact value of $\sin\frac{5\pi}{24}\sin\frac{\pi}{24}$, showing all necessary working.	2

Question 7 (11 marks)**Marks**

- (a) Sketch $y = 3 \sin^{-1} 2x$. **2**
- (b) Find the exact value of $\cos 255^\circ$. **2**
- (c) (i) Prove that $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos 2\theta$. **2**
- (ii) Hence, or otherwise, show that $\tan^2 \frac{\pi}{8} = 3 - 2\sqrt{2}$. **3**

Question 8 (9 marks)**Marks**

- (a) A curve C is given by the parametric equations

3

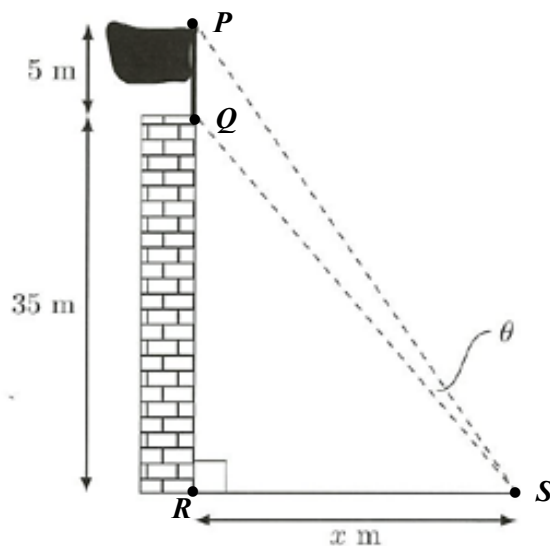
$$x = 2t^2 - 1 \text{ and } y = 3(t + 1)$$

Determine the points of intersection between C and the straight line with equation $3x - 4y = 3$.

- (b) Prove $\frac{\sin 3\theta + \sin \theta}{\cos 3\theta + \cos \theta} = \tan(2\theta)$.

3

- (c) A 5-metre flag stands on top of a 35-metre tall building as shown in the diagram below.

3

The flag subtends an angle of θ with the eye of an observer on the ground x metres away from the building. Show that

$$\theta = \arctan\left(\frac{5x}{x^2 + 1400}\right).$$

END OF ASSESSMENT TASK

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Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET**Measurement****Length**

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

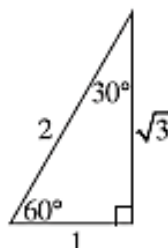
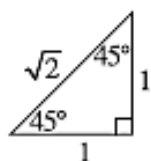
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

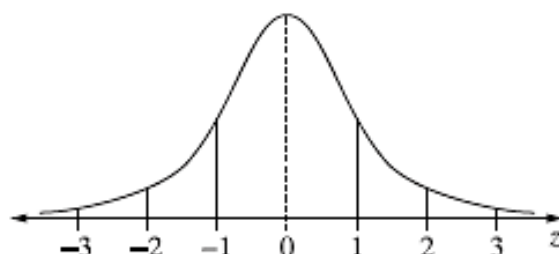
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^nC_x p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1+[f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x_1\underline{i} + y_1\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z &= a + ib = r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

Mrs Cooper
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Name: *solutions*

Teacher:



Pymble Ladies' College

MATHEMATICS EXTENSION 1

Year 11 ASSESSMENT TASK 1

Term 2 2023

Answer Booklet

Time allowed: 55 minutes

Total marks: 33

Section I 5 marks

Section II 28 marks

Instructions

- Put your Name and Teacher's name on this Booklet.
- For Questions 1-5 use the Multiple-Choice Answer Sheet.
- Answer Questions 6, 7 and 8 in the spaces allocated for each question as these spaces provide guidance for the expected length of your response.
- Write using black or blue pen. Black is preferred.
- NESA approved calculators may be used.
- Reference sheets have been provided.

Mark	/33
Rank	/
Highest Mark	/33

Assessment Task 2

SECTION II

28 Marks

Answer all questions in the spaces provided.

Extra writing space is provided at the end of this booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (10 marks)

Marks

(a)

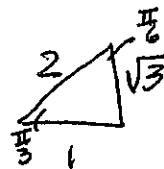
(i)

$$\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = A$$

$$\cos A = -\frac{\sqrt{3}}{2}$$

$$A = \pi - \frac{\pi}{6} \checkmark$$

$$A = \boxed{\frac{5\pi}{6}}$$



1

☐ 1 $A = \pi - \frac{\pi}{6}$

(ii)

$$\cos(\tan^{-1}(-1)) = \cos A$$

$$A = \tan^{-1}(-1) = \cos\left(-\frac{\pi}{4}\right)$$

$$\tan A = -1$$

$$A = -\frac{\pi}{4}$$

$$= \cos \frac{\pi}{4}$$

$$= \boxed{\frac{1}{\sqrt{2}}}$$

2

☐ reaches $\tan A = -1$ or equivalent

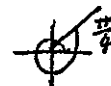
☐ 2 correct soln

(iii)

$$\sin^{-1}\left(\sin \frac{9\pi}{4}\right)$$

$$= \sin^{-1}\left(\sin \frac{\pi}{4}\right)$$

$$= \boxed{\frac{\pi}{4}}$$



2

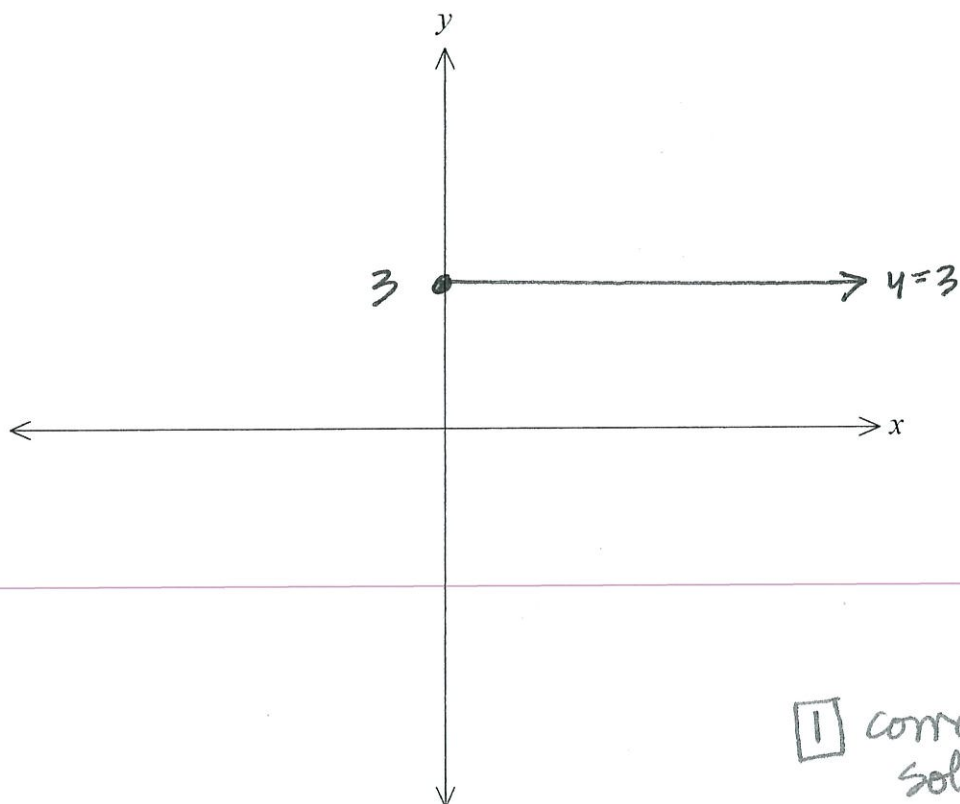
☐ equates $\frac{9\pi}{4}$ with $\frac{\pi}{4}$

☐ 2 correct soln

Question 6 continues next page

(b)

1



$x = p^2$
 $y = 3$
 but $p^2 \geq 0$
 $\therefore x \geq 0$
 Domain: $x \geq 0$

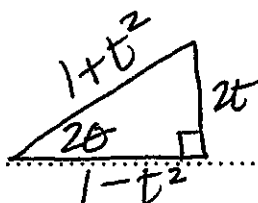
eqn is $y = 3$

Question 6 continued.

Marks

(c)

$$t = \tan \theta$$



2

$$\frac{\cot 2\theta}{1 - \csc 2\theta} = \frac{1-t^2}{2t} \times \frac{2t}{2t}$$

correct subst

[1] $\frac{1-t^2}{2t}$

$$= \frac{1-t^2}{2t - 1-t^2}$$

$$= \frac{1-t^2}{-(t^2 - 2t + 1)}$$

[2] correct soln

$$= \frac{(1+t)(1-t)}{-(t-1)^2}$$

$$= \frac{(t+1)(t-1)}{(t-1)^2}$$

$$= \frac{t+1}{t-1}$$

Question 6 continues next page

Question 6 continued.

Marks

(d) $\sin \frac{5\pi}{24} \cdot \sin \frac{\pi}{24} = \frac{1}{2} \left[\cos \left(\frac{5\pi}{24} - \frac{\pi}{24} \right) - \cos \left(\frac{5\pi}{24} + \frac{\pi}{24} \right) \right]$ 2

$$= \frac{1}{2} \left[\cos \frac{\pi}{6} - \cos \frac{\pi}{4} \right]$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{1}{4} - \frac{1}{2\sqrt{2}}$$

attempts to
[1] use
product to
sum

[2] $\frac{1}{2} \left(\frac{1}{2} - \frac{1}{\sqrt{2}} \right)$
full marks
(rationalisation
not
required)

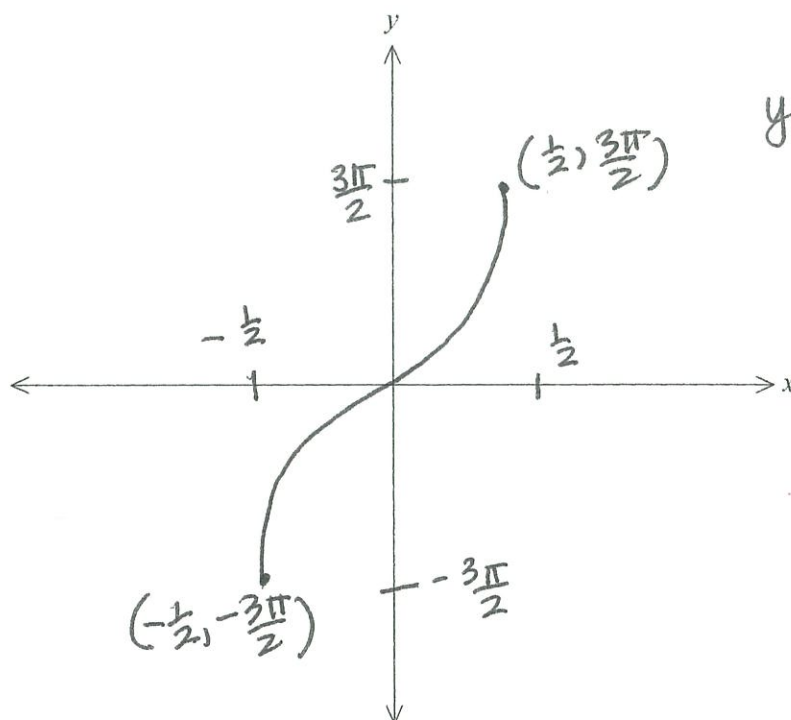
End of Question 6

Question 7 (9 marks)

Marks

(a)

2



$$y = 3 \sin^{-1} 2x$$

1 correct range or domain

2 correct graph.

$$y = 3 \sin^{-1} 2x$$

$$\frac{y}{3} = \sin^{-1} 2x$$

$$D: -1 \leq 2x \leq 1$$

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$R: -\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$$

$$-3\frac{\pi}{2} \leq y \leq 3\frac{\pi}{2}$$

(b)

2

$$\cos 255^\circ = \cos (210^\circ + 45^\circ)$$

$$= \cos 210^\circ \cos 45^\circ - \sin 210^\circ \sin 45^\circ$$

$$= -\cos 30^\circ \cos 45^\circ - (-\sin 30^\circ) \sin 45^\circ$$

$$= -\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}}$$

1 uses $\cos(A+B)$ expands correctly

2 correct soln (rationalisation not required)

$$-\frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}}$$

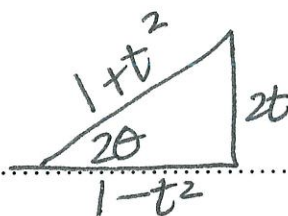
2 marks

Question 7 continues next page

(c)

(i)

let $t = \tan \theta$



2

$$\text{LHS} = \frac{1-t^2}{1+t^2}$$

$$\text{RHS} = \frac{1-t^2}{1+t^2} \quad (\text{from } \Delta)$$

$$= \text{LHS as required}$$

1 correct ~~soln~~
 set up for
 t-formula
 and
 applies t-formula
 to either LHS
 or RHS correctly

2 correct soln

$$\text{LHS} = \frac{1 - \frac{\sin^2 \theta}{\cos^2 \theta}}{1 + \frac{\sin^2 \theta}{\cos^2 \theta}}$$



or

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta}$$

$$= \frac{\cos 2\theta}{1}$$

$$= \text{RHS as required}$$

1 converts
 to $\sin \theta$ and
 $\cos \theta$

or

correct use of one
 identity

$$\cos^2 \theta - \sin^2 \theta = \cos 2\theta$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

2 correct
 soln

(ii)

$$\text{let } \theta = \frac{\pi}{8} \quad \therefore 2\theta = \frac{\pi}{4}$$

$$\text{by (i)} \quad \frac{1 - \tan^2 \frac{\pi}{8}}{1 + \tan^2 \frac{\pi}{8}} = \cos \frac{\pi}{4}$$

$$\text{let } \tan \theta = t$$

$$\therefore \frac{1 - t^2}{1 + t^2} = \frac{1}{\sqrt{2}}$$

$$\sqrt{2} - \sqrt{2}t^2 = 1 + t^2$$

$$\sqrt{2} - 1 = t^2 + \sqrt{2}t^2$$

$$\sqrt{2} - 1 = t^2(1 + \sqrt{2})$$

$$\therefore t^2 = \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1}$$

$$t^2 = \frac{2 - 2\sqrt{2} + 1}{2 - 1}$$

$$t^2 = 3 - 2\sqrt{2} \quad \text{as required}$$

3

$$\boxed{1} \text{ reaches}$$

$$\frac{1 - \tan^2 \frac{\pi}{8}}{1 + \tan^2 \frac{\pi}{8}} = \cos \frac{\pi}{4}$$

gets

$$\boxed{2} \quad t^2 = \frac{\sqrt{2} - 1}{\sqrt{2} + 1}$$

$\boxed{3}$ rationalise
correctly
to

$$t^2 = 3 - 2\sqrt{2}$$

Question 8 (9 marks)

Marks

(a)

$$x = 2t^2 - 1 \quad \text{--- [1]} \quad y = 3(t+1)$$

$$t+1 = \frac{y}{3}$$

$$t = \frac{y}{3} - 1 \quad \text{--- [2]}$$

$$\text{[2]} \rightarrow \text{[1]} \quad x = 2\left(\frac{y}{3} - 1\right)^2 - 1$$

$$x = 2\left(\frac{y^2}{9} - \frac{2y}{3} + 1\right) - 1$$

$$x = \frac{2y^2}{9} - \frac{4y}{3} + 2 - 1$$

$$x = \frac{2y^2}{9} - \frac{4y}{3} + 1 \quad \text{--- [3]} \quad \text{is C}$$

now $3x - 4y = 3$

$$3x = 3 + 4y$$

$$x = 1 + \frac{4y}{3} \quad \text{--- [4]}$$

for pt of intersection of [3] + [4]

[4] $\rightarrow$ [3]

$$1 + \frac{4y}{3} = \frac{2y^2}{9} - \frac{4y}{3} + 1$$

[x9]

$$12y = 2y^2 - 12y$$

$$0 = 2y^2 - 24y$$

$$0 = y^2 - 12y$$

$$0 = y(y-12)$$

$$\rightarrow \text{[7]} \quad \left. \begin{array}{l} \therefore y=0 \\ x=1+\frac{4}{3}(0) \\ x=0 \end{array} \right\} \quad \vee \quad \left. \begin{array}{l} y=12 \\ x=1+\frac{4}{3}(12) \\ x=17 \end{array} \right\}$$

$$\therefore (0,0) \quad \text{and} \quad (12,17)$$

3

[1] subst t
to
 $x = 2\left(\frac{y}{3} - 1\right)^2 - 1$

[2] isolates
variable
from
 $3x - 4y = 3$
and subs
into C eqn

[3] correct
solns

Question 8 continues next page

(b)

$$\text{LHS} = \frac{\sin 3\theta + \sin \theta}{\cos 3\theta + \cos \theta}$$

$$A+B=3\theta$$

$$A-B=\theta$$

$$2A=4\theta$$

$$\therefore A=2\theta, B=\theta$$

$$= \frac{\sin(2\theta+\theta) + \sin(2\theta-\theta)}{\cos(2\theta+\theta) + \cos(2\theta-\theta)}$$

$$= \frac{2 \sin 2\theta \cdot \cos \theta}{2 \cos 2\theta \cdot \cos \theta}$$

$$= \tan 2\theta$$

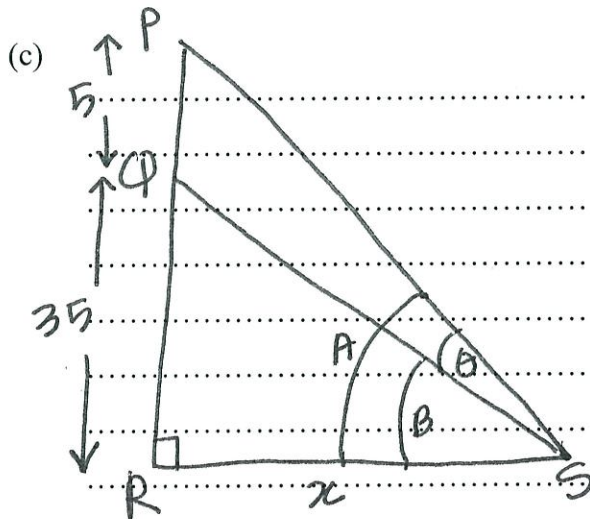
$$= \text{RHS as required}$$

3

① attempts to use sums to products

② reaches either
 $2 \sin 2\theta \cdot \cos \theta$
 or
 $2 \cos 2\theta \cdot \cos \theta$

③ correct soln



$$\text{let } A = \angle PSR$$

$$B = \angle QSR$$

$$\therefore \theta = A - B$$

$$\tan A = \frac{40}{x} \quad \tan B = \frac{35}{x}$$

$$\tan \theta = \tan (A - B)$$

$$= \frac{\tan A - \tan B}{1 + \tan A \cdot \tan B}$$

$$= \frac{\frac{40}{x} - \frac{35}{x}}{1 + \frac{40}{x} \cdot \frac{35}{x}} \times \frac{x^2}{x^2}$$

$$= \frac{40x - 35x}{x^2 + 1400}$$

$$\tan \theta = \frac{5x}{x^2 + 1400}$$

$$\therefore \theta = \tan^{-1} \left(\frac{5x}{x^2 + 1400} \right)$$

$$\text{or } \theta = \arctan \left(\frac{5x}{x^2 + 1400} \right)$$

1 sets up
 $\theta = A - B$ or
equivalent

(or)

$$\tan A = \frac{40}{x}$$

(or)

$$\tan B = \frac{35}{x}$$

(or)

attempts to use
 $\tan(A - B)$ expansion

2

uses $\tan(A - B)$
expansion
and subst.
correctly

3 reaches

$$\theta = \tan^{-1} \left(\frac{5x}{x^2 + 1400} \right)$$

(or)

$$\theta = \arctan \left(\frac{5x}{x^2 + 1400} \right)$$

END OF ASSESSMENT TASK